

# Linear Algebra and Calculus Lecture 6

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Fall 2025

## Vectors

A **column vector** (or simply **vector**) is a matrix with only one column. For example

$$\begin{bmatrix} 5 \\ 6 \end{bmatrix}, \begin{bmatrix} \sqrt{2} \\ \frac{1}{3} \end{bmatrix}, \begin{bmatrix} \pi \\ 1 \end{bmatrix}.$$

We will write  $\mathbb{R}^2$  for the set of  $2 \times 1$  matrices or column vectors with two (real number) entries. Elements will typically be written as  $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ . Note that the addition of vectors is properly defined:

$$v + w = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix},$$

which can be represented by the parallelogram rule for addition. It is also defined the multiplication by a scalar  $c$ :

$$c \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{bmatrix}.$$

More generally for an integer  $n \geq 1$  we can consider the set  $\mathbb{R}^n$  of  $n \times 1$  matrices with real numbers as entries. As before there is an addition on  $\mathbb{R}^n$  and a scalar multiplication  $c \cdot v$  for  $v \in \mathbb{R}^n$  and  $c$  a scalar. These satisfy the following algebraic properties

- |                                        |                         |
|----------------------------------------|-------------------------|
| 1. $u + v = \rule{1cm}{0.4pt}$         | 5. $c(u + v) = cu + cv$ |
| 2. $u + (v + w) = \rule{1.5cm}{0.4pt}$ | 6. $(c + d)u = cu + du$ |
| 3. $v + 0 = 0 + v = v$                 | 7. $c(du) = (cd)u$      |
| 4. $u + (-u) = -u + u = 0$             | 8. $1u = u$             |

## Linear combinations

Let  $v_1, \dots, v_p$  be elements of  $\mathbb{R}^n$ . Then we say an element  $y \in \mathbb{R}^n$  is a linear combination of  $v_1, \dots, v_p$  if there are scalars called coefficients  $c_1, \dots, c_p \in \mathbb{R}$  such that  $y = \rule{1.5cm}{0.4pt}$ . For example,

$$\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

**Question 1.** Let  $a_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$ ,  $a_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$  and  $b = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$ . Can  $b$  be generated (or written) as a linear combination of  $a_1$  and  $a_2$ ?

This amounts to finding the weights (if they exist) of such linear combination. That is, we want to find real numbers  $x_1, x_2$  such that

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$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

which is the same as

$$\begin{bmatrix} x_1 \\ -2x_1 \\ -5x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

and

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Two vectors are equal if their entries are equal, so this holds true if

$$\begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \\ -5x_1 + 6x_2 &= -3 \end{aligned}$$

Which is a linear system of equations! We can write the augmented matrix and row reduce it:

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The solution is \_\_\_\_\_. Hence  $b$  is a linear combination of  $a_1$  and  $a_2$ .

**Exercise.** Plug the solution of  $x_1$  and  $x_2$  into the linear combination equation and check that the equality holds.

**Proposition.** A vector equation

$$x_1 a_1 + x_2 a_2 + \cdots + x_n a_n = b$$

has the same solution set as the linear system whose augmented matrix is

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In particular,  $b$  can be generated by a linear combination of  $a_1, \dots, a_n$  if and only if there exists a solution to the linear system corresponding to the previous matrix.

## Span

**Definition.** The **span**,  $\text{Span}\{v_1, \dots, v_p\}$ , of a set of vectors  $v_1, \dots, v_p \in \mathbb{R}^n$  is the set of vectors  $w \in \mathbb{R}^n$  that are a \_\_\_\_\_ of  $v_1, \dots, v_p$ , that is, that can be written in the form

$$c_1v_1 + c_2v_2 + \dots + c_pv_p$$

with  $c_1, \dots, c_p$  scalars.

Asking whether a vector  $b$  is in  $\text{Span}\{v_1, \dots, v_p\}$  amounts to asking whether the vector equation

\_\_\_\_\_ has a solution, or, equivalently, asking whether the linear system with augmented matrix  $\begin{bmatrix} \text{---} & \text{---} & \text{---} & \text{---} \end{bmatrix}$  has a solution.

## The Matrix Equation $Ax = b$

Recalling the previous lecture, it is possible to linearly combine the columns of a matrix. Let us formalize that.

**Definition.** If  $A$  is an  $m \times n$  matrix, with columns  $a_1, \dots, a_n$ , and if  $x \in \mathbb{R}^n$ , then the product of  $A$  and  $x$ , denoted by  $Ax$ , is the linear combination of the columns of  $A$  using the corresponding entries in  $x$  as weights; that is

$$Ax = \text{---} = x_1a_1 + x_2a_2 + \dots + x_na_n.$$

**Example.** For  $v_1, v_2, v_3 \in \mathbb{R}^n$ , we can write the linear combination  $3v_1 - 5v_2 + 7v_3$  as a matrix times a vector:

$$3v_1 - 5v_2 + 7v_3 = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \text{---} = Ax.$$

The equation  $Ax = b$  is called the **matrix equation**.

**Theorem.** If  $A$  is an  $m \times n$  matrix, with columns  $a_1, \dots, a_n$  and if  $b \in \mathbb{R}^m$ , the matrix equation

$$Ax = b$$

has the same solution set as the vector equation

\_\_\_\_\_ which in turn, has the same solution set as the system of linear equations whose augmented matrix is

\_\_\_\_\_ It follows from this theorem, that the equation  $Ax = b$  has a solution if and only if  $b$  is a linear combination of the columns of  $A$ .

**Example.** Let  $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$  and  $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ . Is the equation  $Ax = b$  consistent for all possible  $b_1, b_2, b_3$ ?

We can row reduce the augmented matrix for  $Ax = b$ :

$$\left[ \begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 7 & 5 & \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2 + 4b_1 \\ 0 & 0 & 0 & \end{array} \right].$$

The third entry in column 4 equals  $b_1 - \frac{1}{2}b_2 + b_3$ . The equation  $Ax = b$  is not consistent for every  $b$  because some choices of  $b$  can make  $b_1 - \frac{1}{2}b_2 + b_3$  non-zero. For it to be consistent it is required that

$$\underline{\hspace{2cm}}.$$

**Theorem.** Let  $A$  be an  $m \times n$  matrix. Then the following statements are logically equivalent. That is, for a particular  $A$ , either they are all true statements or they are all false.

1. For each  $b \in \mathbb{R}^m$ , the equation  $\underline{\hspace{2cm}}$  has a solution.
2. Each  $b \in \mathbb{R}^m$  is a linear combination of  $\underline{\hspace{2cm}}$ .
3. The columns of  $A$  span  $\underline{\hspace{2cm}}$ .
4.  $A$  has a pivot position in every  $\underline{\hspace{2cm}}$ .

**Exercise.** Compute the matrix-vector product using the row-vector rule for the following matrix and vector:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix}.$$

**Theorem.** If  $A$  is an  $m \times n$  matrix,  $u$  and  $v$  are vectors in  $\mathbb{R}^n$ , and  $c$  is a scalar, then:

1.  $A(u + v) = \underline{\hspace{2cm}};$
2.  $A(cu) = \underline{\hspace{2cm}}.$